Cyber Security & Networking Fundamentals

Comprehensive Study Notes & Technical Reference Guide

1. Introduction to Cyber Security & IoT

IoT (Internet of Things): Refers to anything connected to the internet.

Examples: CCTV cameras, mobile devices, laptops, smart home devices, etc.

2. Computer Networks & Protocols

What is a Network? Networking enables computers to communicate and share resources through interconnected systems using standardized protocols and addressing schemes.

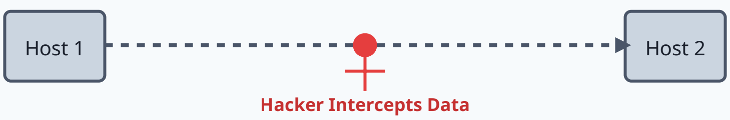
Protocol Definition: Protocols are sets of rules that the internet and devices follow in order to establish communication and transfer data effectively.

Role of Networking in Cyber Security

Networking concepts are crucial in cybersecurity for the following applications:

- **Packet Sniffing & Analysis:** Intercepting and decoding network traffic to identify vulnerabilities and sensitive data transmissions.
- **Protocol Exploitation:** Understanding weaknesses in the TCP/IP stack to execute attacks such as session hijacking or DNS poisoning.
- **Network Reconnaissance:** Information gathering about a target. Mapping target infrastructure using tools like *Nmap* to discover active hosts, open ports, and running services.
- **Firewall Evasion:** Bypassing security controls by manipulating packet headers and understanding filtering mechanisms.

DIAGRAM: PACKET INTERCEPTION ATTACK



3. Types of Networks

- **LAN (Local Area Network):** Connects devices within a limited geographical area like an office building or campus. Highly secure due to its small range.
- **WAN (Wide Area Network):** Spans large distances across cities, countries, or continents, connecting multiple LANs together.
- **MAN (Metropolitan Area Network):** Covers a city or large town area. Larger than a LAN but smaller than a WAN.

- **PAN (Personal Area Network):** Connects devices within a personal workspace, such as smartphones, laptops, and peripherals.

Fiber Optic Cables: Cables installed underwater (submarine cables) and underground, connecting countries and continents for high-speed data transfer and global communication.

4. Network Devices

Device	Function / Description	Key Feature
Router	Forwards data packets between different networks using IP addresses.	Routes data safely to other LANs
Switch	Connects devices on the same network using MAC addresses. Discovers which MAC address data belongs to and transfers it.	Intelligent local frame distribution
Hub	Broadcasts data to all connected devices without using MAC addresses.	**Obsolete technology** (not used today)
Firewall	Filters network traffic based on security rules, blocking unauthorized access to protect data from hackers.	Blocks or allows traffic based on rules
Modem	Converts digital signals to analog (and vice versa) for internet connectivity.	Modulation / Demodulation
Access Point	Provides wireless network connectivity (Wi-Fi) to devices.	Wireless connection gateway

5. IP Address vs MAC Address

IP Address (Internet Protocol)

Definition: Logical address assigned by the network.

Key Properties:

- Changes when a device moves to a different network.
- Assigned by network administrator or via **DHCP** (Dynamic Host Configuration Protocol).
- Used for routing packets across networks.
- Identifies device location on the internet.

Versions: IPv4 and IPv6

MAC Address (Media Access Control)

Definition: Physical address burned directly into device hardware (NIC).

Key Properties:

- Permanent and unique to every device.
- Never changes.
- Used for local network communication within the same LAN segment.

6. IPv4 vs IPv6 Comparison

Property	IPv4	IPv6
Format	32-bit address, dotted decimal notation `192.168.0.1`	128-bit address, hexadecimal notation `2001:0db8::1`
Structure	Divided into 4 octets (8 bits each)	Divided into 8 groups (16 bits each)
Address Space	$2^{32} \approx$ 4.3 Billion addresses	$2^{128} \approx 3.4 \times 10^{38}$ addresses
Current Adoption	Widely used globally today.	Newer version created because world population & address demand exceeded IPv4 capacity.

7. Public vs Private IP Addresses

Public IP	Private IP
Definition: Globally unique address routable on the public internet.	**Definition:** Non-routable address used strictly within internal/private networks.
Example: `8.8.8.8` (Google DNS)	**Example:** `192.168.0.12`
Use Case: Public servers, websites, web-facing infrastructure. Communicates directly with other public IPs.	**Use Case:** Home networks, corporate LANs, internal smart devices. Communicates locally.

NAT (Network Address Translation): Private IP addresses are translated into a Public IP address by a router using NAT when accessing external internet resources.

Reserved Private IP Address Ranges

- **Class A:** `10.0.0.0` to `10.255.255.255`
- **Class B:** `172.16.0.0` to `172.31.255.255`
- **Class C:** `192.168.0.0` to `192.168.255.255`

8. Static vs Dynamic IP Assignment

Static IP	Dynamic IP
Definition: Manually assigned fixed IP address that does not change over time.	**Definition:** Automatically assigned IP address provided dynamically by a DHCP server.
Characteristics: • Assigned manually by network admin • Used for servers, printers, critical infrastructure • More expensive to acquire / maintain	**Characteristics:** • Changes periodically or on reconnecting • Managed automatically via DHCP protocol • Used for home networks, smartphones, PCs

9. IP Address Classes (Classful Addressing)

Classful addressing is the original method of dividing IPv4 addresses into classes based on the value of the first octet.

Class	First Octet Range	Network Bits	Host Bits	Usable Hosts / Purpose
Class A	1 – 126	8 bits	24 bits	16 Million hosts (Large networks)
Class B	128 – 191	16 bits	16 bits	65,000 hosts (Medium networks)
Class C	192 – 223	24 bits	8 bits	254 hosts (Small/Home networks)
Class D	224 – 239	Reserved for Multicast		Multicast group communication
Class E	240 – 255	Reserved		Experimental / Research only

Example Structure: In Class C address `192.168.11.10`, the first three octets (`192.168.11`) represent the **Network Portion**, and the last octet (`10`) represents the **Host Portion**.

10. CIDR Notation & Subnetting

CIDR (Classless Inter-Domain Routing): A flexible method to express IP addresses and their network size in a single line notation.

CIDR Notation	Subnet Mask	Usable Hosts	Network Size Scale
`/8`	255.0.0.0	16,777,214	Very Large Network
`/16`	255.255.0.0	65,534	Medium Network
`/24`	255.255.255.0	254	Small Network / Standard LAN
`/26`	255.255.255.192	62	Subnetted Segment
`/30`	255.255.255.252	2	Point-to-Point Link

What is Subnetting?

Subnetting is the practice of dividing a single large network into smaller, logically isolated, and manageable network segments.

Practical Example: Dividing a single network `192.168.1.0/24` into distinct subnets (`/26`) allows an organization to assign separate IP ranges to different departments (e.g., Developers, HR, Sales, Management) for enhanced security and traffic management.

Subnet A	Subnet B	Subnet C	Subnet D
`192.168.1.0/26`	`192.168.1.64/26`	`192.168.1.128/26`	`192.168.1.192/26`
Hosts: 62	Hosts: 62	Hosts: 62	Hosts: 62

Computer Networking & Protocols Guide

Complete Reference Notes — NAT, OSI, TCP/IP, Port Mapping & Handshakes

1. Network Address Translation (NAT)

Definition & Overview

NAT (Network Address Translation) allows private IP addresses within a Local Area Network (LAN) to communicate with external networks (the Internet) by translating them into public IP addresses.

Key Note: Subnet configurations may allocate specific capacities (e.g., each department receiving 62 usable IPs).

Translation Workflow Example

1. Private Device (192.168.1.10): Initiates outbound traffic to a remote target.

2. Router (NAT Engine): Replaces the private source IP with its registered Public IP address.

3. Public Internet (203.0.113.45): Routes packet across public servers (e.g., Google servers).

4. Response Handling (Inbound): Router receives incoming response, translates Public IP back to internal Private IP, and routes to original device.

Why NAT Exists

- **IP Address Conservation:** Solves IPv4 exhaustion by allowing thousands of devices to share a single public IP.
- **Enhanced Security:** Keeps internal IP structures hidden from external scanning and threats.
- **Network Flexibility:** Enables internal networks to change topologies without altering external public addresses.
- **Privacy:** Private IP addresses are strictly local and never advertised directly over the public Internet.

2. Protocols & Port Architecture

Protocols

Ports

Defined sets of rules and standards governing end-to-end data communication across network devices.

Examples: HTTP, HTTPS, FTP, DNS, SSH, SMTP, SSL/TLS.

Logical endpoints ("doors") on a host facilitating specific multi-app communications over a single IP connection.

Total Range: 0 to 65535 (65,536 total ports).

Port Categories & Ranges

RANGE	CLASSIFICATION	DESCRIPTION
`0 - 1023`	**Well-Known Ports**	Reserved for core system services and ubiquitous protocols.
`1024 - 49151`	**Registered Ports**	Assigned by IANA for specific vendor applications/services.
`49152 - 65535`	**Dynamic / Private Ports**	Used client-side as ephemeral ports during active sessions.

Standard Protocol & Port Map

PROTOCOL	PORT NUMBER	PRIMARY FUNCTION
HTTP	80	Unencrypted Web Traffic
HTTPS	443	Secure Encrypted Web Traffic (SSL/TLS)
FTP	21	File Transfer Protocol Control
SSH	22	Secure Shell Remote Access
DNS	53	Domain Name System Resolution

3. OSI Model vs. TCP/IP Model

Architectural Comparison

The **OSI Model** serves as a theoretical 7-layer conceptual framework. The operational standard implemented across the global internet is the streamlined 4-layer **TCP/IP Model**.

OSI LAYER	MAPPED TCP/IP LAYER	KEY RESPONSIBILITIES & PROTOCOLS
7. Application 6.	**Application Layer**	

OSI LAYER	MAPPED TCP/IP LAYER	KEY RESPONSIBILITIES & PROTOCOLS
Presentation 5. Session		User interfaces, network services, encryption, session maintenance. *Protocols: HTTP, HTTPS, DNS, FTP, SMTP*
4. Transport	**Transport Layer**	End-to-end reliability, flow control, port addressing. *Protocols: TCP, UDP*
3. Network	**Internet Layer**	Logical IP addressing and packet routing choices. *Protocols: IP, ICMP, ARP*
2. Data Link 1. Physical	**Network Access Layer**	Hardware MAC addressing, framing, physical bit transmission. *Protocols: Ethernet, Wi-Fi, Hardware*

4. Data Encapsulation & Protocol Mechanics

Data Encapsulation Sequence

```
[ Application ] → Data
↓
[ Transport ] → Segment (Adds TCP/UDP Header)
↓
[ Network ] → Packet (Adds IP Header)
↓
[ Data Link ] → Frame (Adds MAC Address Header/Trailer)
↓
[ Physical ] → Bits (Physical Signals / Wire)
```

Encapsulation wraps data with operational headers at each descending layer. *Decapsulation* strip-reads headers in reverse at the destination host.

TCP vs. UDP Protocol Breakdown

FEATURE	TCP (TRANSMISSION CONTROL PROTOCOL)	UDP (USER DATAGRAM PROTOCOL)
Connection Type	Connection-Oriented	Connectionless
Speed & Overhead	Slower (Header/Handshake Overhead)	Faster (Minimal Overhead)
Reliability	Guaranteed Delivery (ACKs)	Best-Effort (No Guarantee)
Error Control	Extensive Error Checking & Retransmission	Basic Checksum Only

FEATURE	TCP (TRANSMISSION CONTROL PROTOCOL)	UDP (USER DATAGRAM PROTOCOL)
Ordering	Strict Sequence Ordering Maintained	No Packet Ordering Guarantee
Primary Use Cases	Web Browsing, Email, File Transfers	Online Gaming, Live Streaming, DNS

5. TCP Handshakes & Connection Control

TCP 3-Way Handshake (Connection Establishment)

```
Client   --- SYN ("Request Connection") --->   Server
Client   <--- SYN-ACK ("Acknowledge & Sync") ---   Server
Client   --- ACK ("Connection Established") --->   Server
```

TCP 4-Way Handshake (Connection Termination)

```
Client   --- FIN ("Done sending data") --->   Server
Client   <--- ACK ("Received termination request") ---   Server
Client   <--- FIN ("Server also done sending") ---   Server
Client   --- ACK ("Closed confirmed") --->   Server
```

Common TCP Control Flags

- `SYN` **Synchronize:** Initiates and negotiates connection parameters.
- `ACK` **Acknowledge:** Confirms successful receipt of transmitted packets.
- `FIN` **Finish:** Requests graceful termination of the connection.
- `RST` **Reset:** Immediately aborts/resets an unstable connection.
- `PSH` **Push:** Forces immediate data delivery to app without buffering.
- `URG` **Urgent:** Marks payload as urgent priority for immediate processing.

Common Protocols Quick Reference

Protocol	`Port`	Purpose
HTTP	80	Unencrypted web traffic • Hyper Text Transfer Protocol Unencrypted mean data transfer in same form **Example** → Hi send in same form → Hi
HTTPS	443	Encrypted web traffic (SSL/TLS) • Hyper text transfer Protocol secure. Encrypted mean data transfer into cipher language
FTP	21	File transfer Protocol (unencrypted)
SFTP	22	Secure file transfer (SSH)
SSH	22	secure shell, remote access
Telnet	23	Remote access (unencrypted, deprecated) Telnet is older version of SSH.
DNS	53	Domain name system • It helps in Domain name resolution.
DHCP	67 – 68	Dynamic IP assignment → automatically assign IP to devices.
SMTP	25	send mail.
POP3	110	Receive email (older)
IMAP	143	Receive email (modern)
RDP	3389	Remote desktop (Windows)

Encryption & Decryption Process

Definition:
converting plain text into ciphertext (encryption) and back (decryption) using decryption keys.

Encryption	Decryption
step 1: Plaintext (Readable data) Encryption key	**step 1:** Ciphertext (encrypted data) * Decryption.

Encryption	Decryption
step 2: Encryption Algorithm (AES, RSA, DES) **step 3:** Ciphertext (unreadable scrambled data) **Result:** Data is now secure and unreadable without key	**step 2:** Decryption Algorithm (reverse of encryption) **step 3:** Plaintext (original readable data) **Result:** Data is recovered and readable again

Symmetric Encryption:

Same key for encryption and decryption (AES, DES)

- Fast, efficient
- Key must be shared securely
- Used for bulk data

Asymmetric Encryption:

Different Key

- slower but secure
- Public key encryption and Private key decrypt
- mostly used in key exchange and digital signature

DNS - How Domain Resolution Works (Domain Name System)

1. User types `google.com` in browser

↓

2. Query sent to Recursive Resolver (ISP DNS)

↓

3. Resolver queries Root Name server (`root .`)

↓

4. Root directs to TLD server (`.com`) (Top level domain)

↓

5. TLD server directs to Authoritative Nameserver

↓

6. Authoritative server returns IP address (142.250.195.46)

↓

7. Browser connects to IP and loads website.